Setup

In [72]:
```
#STANDARD IMPORTS
import sympy as sp
import numpy as np
import os
import matplotlib.pyplot as plt
### JUPYTER NOTEBOOK SETTINGS ########################################
#Plot all figures in full-size cells, no scroll bars
%matplotlib inline
#Disable Python Warning Output
#(NOTE: Only for production, comment out for debugging)
import warnings
warnings.filterwarnings('ignore')
### PLOTTING DEFAULTS BOILERPLATE (OPTIONAL) #########################
#SET DEFAULT FIGURE APPERANCE
import seaborn as sns #Fancy plotting package
#No Background fill, legend font scale, frame on legend
sns.set(style='whitegrid', font_scale=1.5, rc={'legend.frameon': True})
#Mark ticks with border on all four sides (overrides 'whitegrid')
sns.set_style('ticks')
#ticks point in
sns.set_style({"xtick.direction": "in","ytick.direction": "in"})
#fix invisible marker bug
sns.set_context(rc={'lines.markeredgewidth': 0.1})
#restore default matplotlib colormap
mplcolors = ['#1f77b4', '#ff7f0e', '#2ca02c', '#d62728', '#9467bd',
'#8c564b', '#e377c2', '#7f7f7f', '#bcbd22', '#17becf']
sns.set_palette(mplcolors)

#Get color cycle for manual colors
colors = sns.color_palette()
#SET MATPLOTLIB DEFAULTS
#(call after seaborn, which changes some defaults)
params = {
#FONT SIZES
'axes.labelsize' : 30, #Axis Labels
'axes.titlesize' : 30, #Title
'font.size' : 28, #Textbox
'xtick.labelsize': 22, #Axis tick labels
'ytick.labelsize': 22, #Axis tick labels
'legend.fontsize': 24, #Legend font size
'font.family' : 'serif',
'font.fantasy' : 'xkcd',
'font.sans-serif': 'Helvetica',
'font.monospace' : 'Courier',
#AXIS PROPERTIES
'axes.titlepad' : 2*6.0, #title spacing from axis
'axes.grid' : True, #grid on plot
'figure.figsize' : (8,8), #square plots
'savefig.bbox' : 'tight', #reduce whitespace in saved figures
#LEGEND PROPERTIES
'legend.framealpha' : 0.5,
'legend.fancybox' : True,
'legend.frameon' : True,
'legend.numpoints' : 1,
'legend.scatterpoints' : 1,
'legend.borderpad' : 0.1,
'legend.borderaxespad' : 0.1,
'legend.handletextpad' : 0.2,
'legend.handlelength' : 1.0,
'legend.labelspacing' : 0,
}
import matplotlib
matplotlib.rcParams.update(params) #update matplotlib defaults, call after
### END OF BOILERPLATE ##################################################
colors = sns.color_palette() #color cycle
```

For each of the plots: Chord length (c=1), Camber line must be plotted as a dotted line, Label upper and lower surface seperately , Chord Line, Use equal axes when plotting

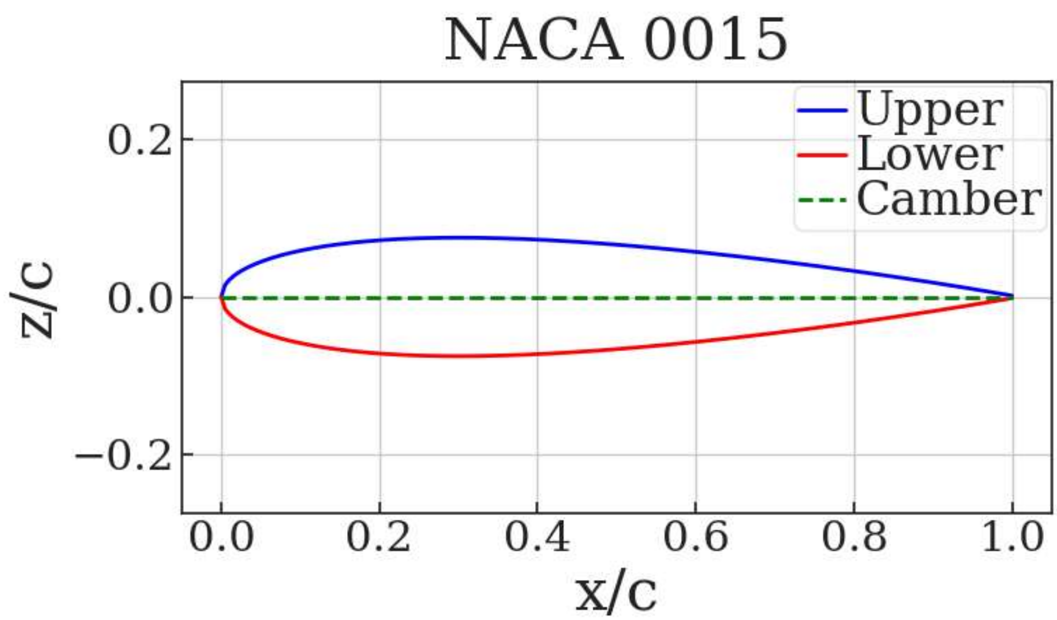
Notes

1.1 Designing Symmetric Airfoils

In [73]:
```
#NACA Code Input
n1=0
```

```
n2=0
n3=1
n4=5

#define variables
c=1 #chord length
maxcam=(n1/100)*c #max camber
maxcamp=(n2/10)*c #max camber position
maxt= (((n3*10)+n4)/100)*c #max thickness
t=maxt/c #thickness


x=np.linspace(0,1,201)

zt=c*(t/0.2)*(
    (0.2969*np.sqrt(x/c))
    -(0.1260*(x/c))
    -(0.3516*(x/c)**2)
    +(0.2843*(x/c)**3)
    -(0.1015*(x/c)**4)
    )


UpperSurface= zt
LowerSurface=-zt

AreaUpper = np.trapz(UpperSurface, x)  # Area under upper surface
AreaLower = np.trapz(LowerSurface, x)  # Area under lower surface

# The chord line is the average of the areas
ChordLine = (AreaUpper - AreaLower) / c
CamberLine = (UpperSurface + LowerSurface) / 2


naca_code = f"NACA {n1}{n2}{n3}{n4}"

# Plotting
plt.figure(figsize=(8, 4))
plt.title(f'{naca_code}')  # Auto-populated title
plt.xlabel("x/c")
plt.ylabel("z/c")
plt.plot(x, UpperSurface, label="Upper", color='blue', linestyle="-", linewidth=2, markersize=8)
plt.plot(x, LowerSurface, label="Lower", color='red', linestyle="-", linewidth=2, markersize=8)
plt.plot(x, CamberLine, label='Camber', color='green', linestyle='--', linewidth=2)
plt.grid(True)
plt.axis('equal')
plt.legend()


#print("Surface Coordinates (x, UpperSurface, LowerSurface):")
#for i in range(len(x)):
#    print(f"x = {x[i]:.4f}, UpperSurface = {UpperSurface[i]:.4f}, LowerSurface = {LowerSurface[i]:.4f}")
```

Out[73]:
```
<matplotlib.legend.Legend at 0x23e146dd3a0>
```

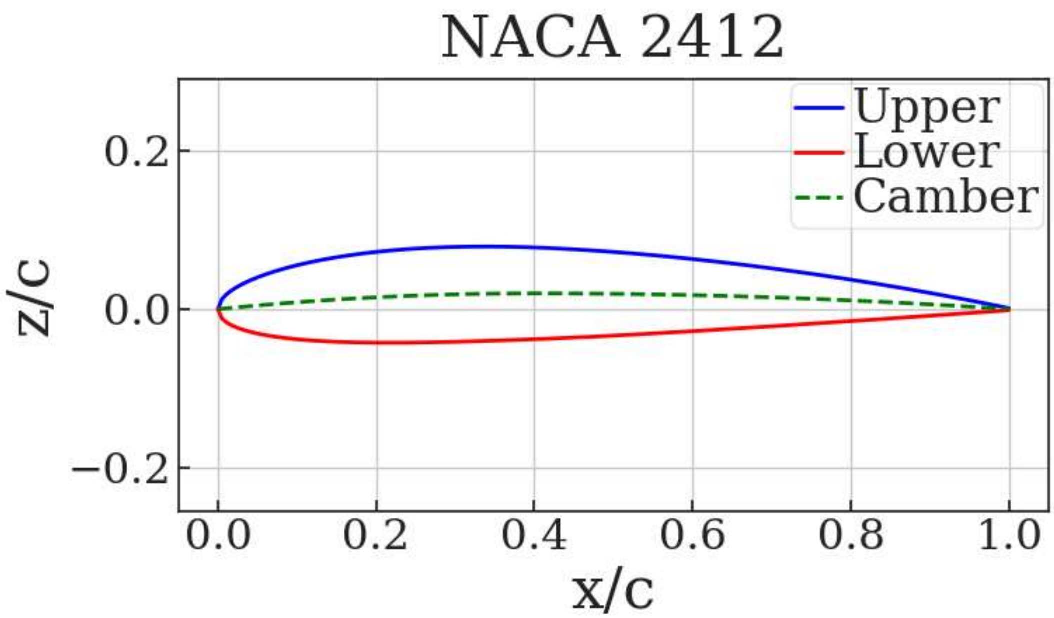


1.2 Designing Cambered Airfoils Airfoil 1

In [74]:
```python
#NACA Code Input
n1=2 #must be nonzero
n2=4 #must be nonzero
n3=1
n4=2

#define variables
c=1 #chord length
maxcam=(n1/100)*c #max camber
maxcamp=(n2/10)*c #max camber position
maxt= (((n3*10)+n4)/100)*c #max thickness
t=maxt/c #thickness
x=np.linspace(0,1,201)

p=maxcamp
m=maxcam

CamberLine = np.zeros_like(x)

# Calculate the CamberLine
for i in range(len(x)):
    if x[i] <= p:  # assumine c=1 (im not sure why it doesnt work when i multiply it by c)
        CamberLine[i] = (m / p**2) * (2 * p * (x[i]) - (x[i])**2)
    else:
        CamberLine[i] = (m / (1 - p)**2) * ((1 - 2 * p) + 2 * p * (x[i]) - (x[i])**2)

zt=c*(t/0.2)*(
    (0.2969*np.sqrt(x/c))
    -(0.1260*(x/c))
    -(0.3516*(x/c)**2)
    +(0.2843*(x/c)**3)
    -(0.1015*(x/c)**4)
    )


UpperSurface= zt + CamberLine
LowerSurface=-zt + CamberLine

naca_code = f"NACA {n1}{n2}{n3}{n4}"

# Plotting
plt.figure(figsize=(8, 4))
plt.title(f'{naca_code}')  # Auto-populated title
plt.xlabel("x/c")
plt.ylabel("z/c")
plt.plot(x, UpperSurface, label="Upper", color='blue', linestyle="-", linewidth=2, markersize=8)
plt.plot(x, LowerSurface, label="Lower", color='red', linestyle="-", linewidth=2, markersize=8)
plt.plot(x, CamberLine, label='Camber', color='green', linestyle='--', linewidth=2)
plt.grid(True)
plt.axis('equal')
```

```
plt.legend()


#print("Surface Coordinates (x, UpperSurface, LowerSurface):")
#for i in range(len(x)):
#    print(f"x = {x[i]:.4f}, UpperSurface = {UpperSurface[i]:.4f}, LowerSurface = {LowerSurface[i]:.4f}")
```

Out[74]: `<matplotlib.legend.Legend at 0x23e14611e80>`

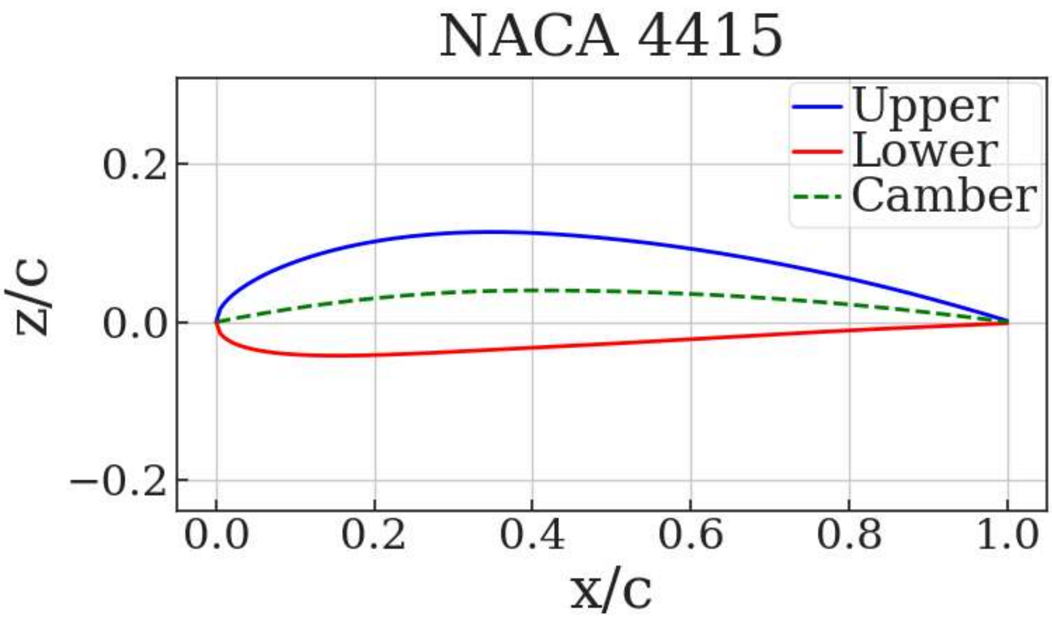


1.2 Designing Cambered Airfoils Airfoil 2

In [75]:
```
#NACA Code Input
n1=4 #must be nonzero
n2=4 #must be nonzero
n3=1
n4=5

#define variables
c=1 #chord length
maxcam=(n1/100)*c #max camber
maxcamp=(n2/10)*c #max camber position
maxt= (((n3*10)+n4)/100)*c #max thickness
t=maxt/c #thickness
x=np.linspace(0,1,201)

p=maxcamp
m=maxcam

CamberLine = np.zeros_like(x)

# Calculate the CamberLine
for i in range(len(x)):
    if x[i] <= p:  # assumine c=1 (im not sure why it doesnt work when i multiply it by c)
        CamberLine[i] = (m / p**2) * (2 * p * (x[i]) - (x[i])**2)
    else:
        CamberLine[i] = (m / (1 - p)**2) * ((1 - 2 * p) + 2 * p * (x[i]) - (x[i])**2)

zt=c*(t/0.2)*(
    (0.2969*np.sqrt(x/c))
    -(0.1260*(x/c))
    -(0.3516*(x/c)**2)
    +(0.2843*(x/c)**3)
    -(0.1015*(x/c)**4)
    )


UpperSurface= zt + CamberLine
LowerSurface=-zt + CamberLine

naca_code = f"NACA {n1}{n2}{n3}{n4}"

# Plotting
plt.figure(figsize=(8, 4))
```

```python
plt.title(f'{naca_code}')  # Auto-populated title
plt.xlabel("x/c")
plt.ylabel("z/c")
plt.plot(x, UpperSurface, label="Upper", color='blue', linestyle="-", linewidth=2, markersize=8)
plt.plot(x, LowerSurface, label="Lower", color='red', linestyle="-", linewidth=2, markersize=8)
plt.plot(x, CamberLine, label='Camber', color='green', linestyle='--', linewidth=2)
plt.grid(True)
plt.axis('equal')
plt.legend()

#print("Surface Coordinates (x, UpperSurface, LowerSurface):")
#for i in range(len(x)):
#    print(f"x = {x[i]:.4f}, UpperSurface = {UpperSurface[i]:.4f}, LowerSurface = {LowerSurface[i]:.4f}")
```

Out[75]: <matplotlib.legend.Legend at 0x23e146dc830>

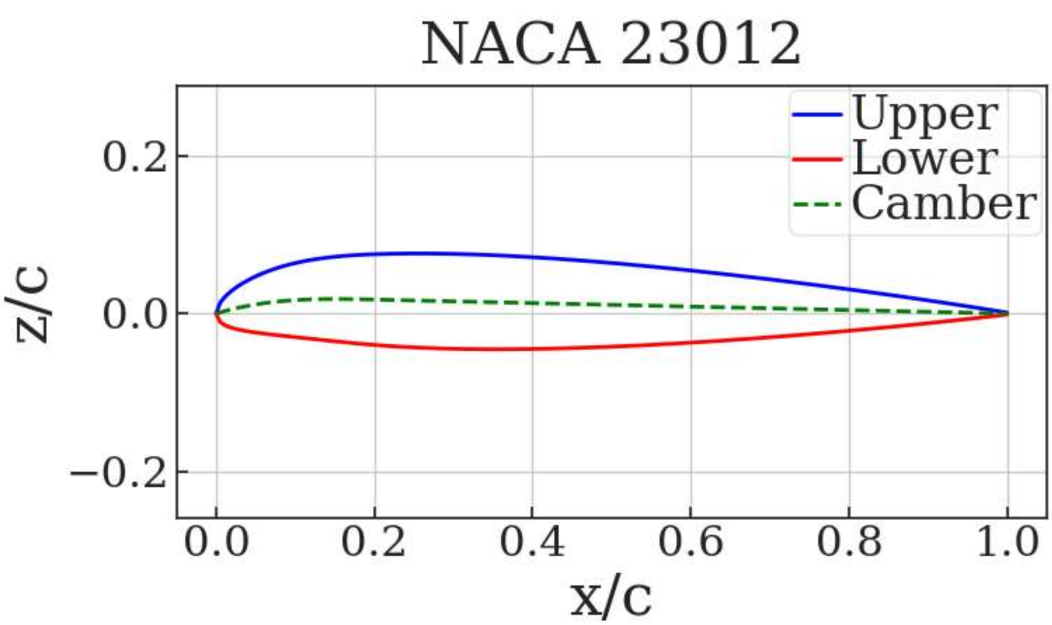


1.3 Designing NACA 5 Digit Airfoil

In [76]:
```python
#NACA Code Input
n1=2 #L
n2=3 #P  (typically 1-5)
n3=0 #S  (0 = simple 1= reflex (in this case always 0))
n4=1 #T
n5=2#T
# r and k1 value storage
rvalues = {1: 0.058, 2: 0.126, 3: 0.2025, 4: 0.29, 5: 0.391}
k1values = {1: 361.4, 2: 51.64, 3: 15.957, 4: 6.643, 5: 3.23}
#selects values based on corresponding n2 number
r=rvalues[n2]
k1=k1values[n2]

c=1 #Chord Length
Cl=n1 #Coefficient of lift
#MaxCamP=n2*0.05 #not used because n2 is used for the r and k1 values
MaxT=(((n4*10)+n5)/100)*c #Maximum Thickness

x=np.linspace(0,c,201)

CamberLine = np.zeros_like(x)

for i in range(len(x)):
    if x[i] <= r:
        CamberLine[i] = (k1 / 6) * (x[i]**3 - 3 * r * x[i]**2 + r**2 * (3 - r) * x[i])
    else:
        CamberLine[i] = (k1 * r**3 / 6) * (1 - x[i])


CamberLineActual=CamberLine*(Cl/2)

zt=c*(MaxT/0.2)*(
    (0.2969*np.sqrt(x/c))
    -(0.1260*(x/c))
    -(0.3516*(x/c)**2)
```

```
    +(0.2843*(x/c)**3)
    -(0.1015*(x/c)**4)
    )

UpperSurface= zt + CamberLine
LowerSurface=-zt + CamberLine


naca_code = f"NACA {n1}{n2}{n3}{n4}{n5}"

# Plotting
plt.figure(figsize=(8, 4))
plt.title(f'{naca_code}')  # Auto-populated title
plt.xlabel("x/c")
plt.ylabel("z/c")
plt.plot(x, UpperSurface, label="Upper", color='blue', linestyle="-", linewidth=2, markersize=8)
plt.plot(x, LowerSurface, label="Lower", color='red', linestyle="-", linewidth=2, markersize=8)
plt.plot(x, CamberLineActual, label='Camber', color='green', linestyle='--', linewidth=2)
plt.grid(True)
plt.axis('equal')
plt.legend()


#print("Surface Coordinates (x, UpperSurface, LowerSurface):")
#for i in range(len(x)):
#    print(f"x = {x[i]:.4f}, UpperSurface = {UpperSurface[i]:.4f}, LowerSurface = {LowerSurface[i]:.4f}")
```

Out[76]: <matplotlib.legend.Legend at 0x23e0ff343e0>



2.0 XFOIL Introduction

In [77]:
```
import pandas as pd
import sys
import pyxfoil

foil = '0012' #NACA airfoil number
naca = True   #allows NACA number input rather than geometry text file
Re   = 0      #Reynolds Number (iAIRFOIL (GEMOETRY FROM EQUATION, NOT FILE)
alfs = [0,4,12]#Angles of Attack to simulated
print("Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...\n")
pyxfoil.GetPolar(foil, naca, alfs, Re, SaveCP=True, quiet=True)

filename=f"C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca0012\\naca0012_polar_Re0.00e+00a0.0-12.
alpha,CL,CD,CDp,CM,Top_Xtr,Bot_Xtr = np.loadtxt(filename, skiprows = 12, unpack = True)
print(alpha,CL,CD,CM)
```

```
Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...

[ 0.  4. 12.] [0.     0.483  1.4395] [0. 0. 0.] [-0.     -0.0056 -0.0164]
```

In [78]:
```
import pandas as pd
import sys
import pyxfoil
```

```
foil = '0012' #NACA airfoil number
naca = True   #allows NACA number input rather than geometry text file
Re   = 436034.9854     #Reynolds Number (iAIRFOIL (GEMOETRY FROM EQUATION, NOT FILE)
alfs = [0,4,12]#Angles of Attack to simulated
print("Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...\n")
pyxfoil.GetPolar(foil, naca, alfs, Re, SaveCP=True, quiet=True)

filename=f"C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca0012\\naca0012_polar_Re4.36e+05a0.0-12.
alpha,CL,CD,CDp,CM,Top_Xtr,Bot_Xtr = np.loadtxt(filename, skiprows = 12, unpack = True)
print(alpha,CL,CD,CM)
```

```
Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...

[ 0.  4. 12.] [0.     0.4973 1.1598] [0.00646 0.00943 0.02794] [-0.     -0.0073  0.0242]
```

In [79]:

```
import pandas as pd
import sys
import pyxfoil

foil = '0012' #NACA airfoil number
naca = True   #allows NACA number input rather than geometry text file
Re   = 1490345.149     #Reynolds Number (iAIRFOIL (GEMOETRY FROM EQUATION, NOT FILE)
alfs = [0,4,12]#Angles of Attack to simulated
print("Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...\n")
pyxfoil.GetPolar(foil, naca, alfs, Re, SaveCP=True, quiet=True)

filename=f"C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca0012\\naca0012_polar_Re1.49e+06a0.0-12.
alpha,CL,CD,CDp,CM,Top_Xtr,Bot_Xtr = np.loadtxt(filename, skiprows = 12, unpack = True)
print(alpha,CL,CD,CM)
```

```
Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...

[ 0.  4. 12.] [0.     0.4315 1.2703] [0.00524 0.00673 0.01695] [-0.      0.0045  0.0099]
```

In [80]:

```
import pandas as pd
import sys
import pyxfoil

foil = '23012' #NACA airfoil number
naca = True   #allows NACA number input rather than geometry text file
Re   = 0      #Reynolds Number (iAIRFOIL (GEMOETRY FROM EQUATION, NOT FILE)
alfs = [0,4,12]#Angles of Attack to simulated
print("Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...\n")
pyxfoil.GetPolar(foil, naca, alfs, Re, SaveCP=True, quiet=True)

filename=f"C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca23012\\naca23012_polar_Re0.00e+00a0.0-1
alpha,CL,CD,CDp,CM,Top_Xtr,Bot_Xtr = np.loadtxt(filename, skiprows = 12, unpack = True)
print(alpha,CL,CD,CM)
```

```
Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...

[ 0.  4. 12.] [0.1376 0.6204 1.5747] [0. 0. 0.] [-0.0116 -0.0175 -0.0311]
```

In [81]:

```
import pandas as pd
import sys
import pyxfoil

foil = '23012' #NACA airfoil number
naca = True   #allows NACA number input rather than geometry text file
Re   = 436034.9854       #Reynolds Number (iAIRFOIL (GEMOETRY FROM EQUATION, NOT FILE)
alfs = [0,4,12]#Angles of Attack to simulated
print("Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...\n")
pyxfoil.GetPolar(foil, naca, alfs, Re, SaveCP=True, quiet=True)

filename=f"C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca23012\\naca23012_polar_Re4.36e+05a0.0-1
alpha,CL,CD,CDp,CM,Top_Xtr,Bot_Xtr = np.loadtxt(filename, skiprows = 12, unpack = True)
print(alpha,CL,CD,CM)
```

```
Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...

[ 0.  4. 12.] [0.1143 0.6432 1.3686] [0.00705 0.0106  0.01945] [-0.0057 -0.0231  0.0022]
```

In [82]:

```
import pandas as pd
import sys
import pyxfoil

foil = '23012' #NACA airfoil number
naca = True   #allows NACA number input rather than geometry text file
Re   = 1490345.149      #Reynolds Number (iAIRFOIL (GEMOETRY FROM EQUATION, NOT FILE)
alfs = [0,4,12]#Angles of Attack to simulated
print("Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...\n")
```

```
pyxfoil.GetPolar(foil, naca, alfs, Re, SaveCP=True, quiet=True)

filename=f"C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca23012\\naca23012_polar_Re1.49e+06a0.0-1
alpha,CL,CD,CDp,CM,Top_Xtr,Bot_Xtr = np.loadtxt(filename, skiprows = 12, unpack = True)
print(alpha,CL,CD,CM)
```

```
Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...

[ 0.  4. 12.] [0.1253 0.5623 1.4203] [0.00612 0.0072  0.01436] [-0.0096 -0.006  -0.0059]
```

In [83]:

```
import pandas as pd
import sys
import pyxfoil

foil = '2612' #NACA airfoil number
naca = True   #allows NACA number input rather than geometry text file
Re   = 0      #Reynolds Number (iAIRFOIL (GEMOETRY FROM EQUATION, NOT FILE)
alfs = [0,4,12]#Angles of Attack to simulated
print("Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...\n")
pyxfoil.GetPolar(foil, naca, alfs, Re, SaveCP=True, quiet=True)

filename=f"C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca2612\\naca2612_polar_Re0.00e+00a0.0-12.
alpha,CL,CD,CDp,CM,Top_Xtr,Bot_Xtr = np.loadtxt(filename, skiprows = 12, unpack = True)
print(alpha,CL,CD,CM)
```

```
Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...

[ 0.  4. 12.] [0.3149 0.7969 1.7469] [0. 0. 0.] [-0.0787 -0.0845 -0.0954]
```

In [84]:

```
import pandas as pd
import sys
import pyxfoil

foil = '2612' #NACA airfoil number
naca = True   #allows NACA number input rather than geometry text file
Re   = 436034.9854     #Reynolds Number (iAIRFOIL (GEMOETRY FROM EQUATION, NOT FILE)
alfs = [0,4,12]#Angles of Attack to simulated
print("Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...\n")
pyxfoil.GetPolar(foil, naca, alfs, Re, SaveCP=True, quiet=True)

filename=f"C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca2612\\naca2612_polar_Re4.36e+05a0.0-12.
alpha,CL,CD,CDp,CM,Top_Xtr,Bot_Xtr = np.loadtxt(filename, skiprows = 12, unpack = True)
print(alpha,CL,CD,CM)
```

```
Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...

[ 0.  4. 12.] [0.2957 0.7238 1.326 ] [0.00719 0.00853 0.02872] [-0.0759 -0.0685 -0.0258]
```

In [85]:

```
import pandas as pd
import sys
import pyxfoil

foil = '2612' #NACA airfoil number
naca = True   #allows NACA number input rather than geometry text file
Re   = 1490345.149      #Reynolds Number (iAIRFOIL (GEMOETRY FROM EQUATION, NOT FILE)
alfs = [0,4,12]#Angles of Attack to simulated
print("Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...\n")
pyxfoil.GetPolar(foil, naca, alfs, Re, SaveCP=True, quiet=True)

filename=f"C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca2612\\naca2612_polar_Re1.49e+06a0.0-12.
alpha,CL,CD,CDp,CM,Top_Xtr,Bot_Xtr = np.loadtxt(filename, skiprows = 12, unpack = True)
print(alpha,CL,CD,CM)
```

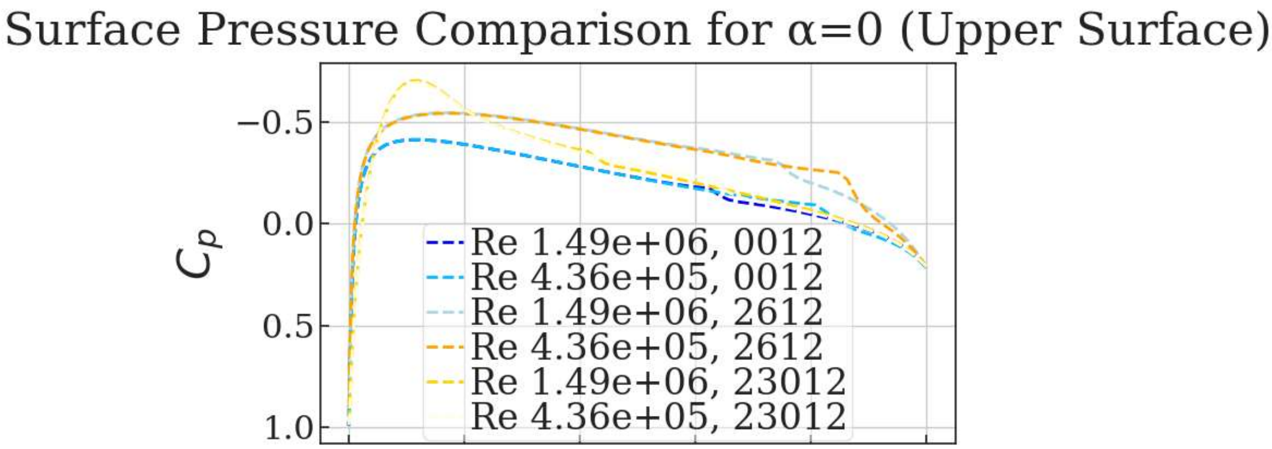
```
Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...

[ 0.  4. 12.] [0.2989 0.7279 1.4681] [0.00506 0.00698 0.01838] [-0.075  -0.0706 -0.0456]
```

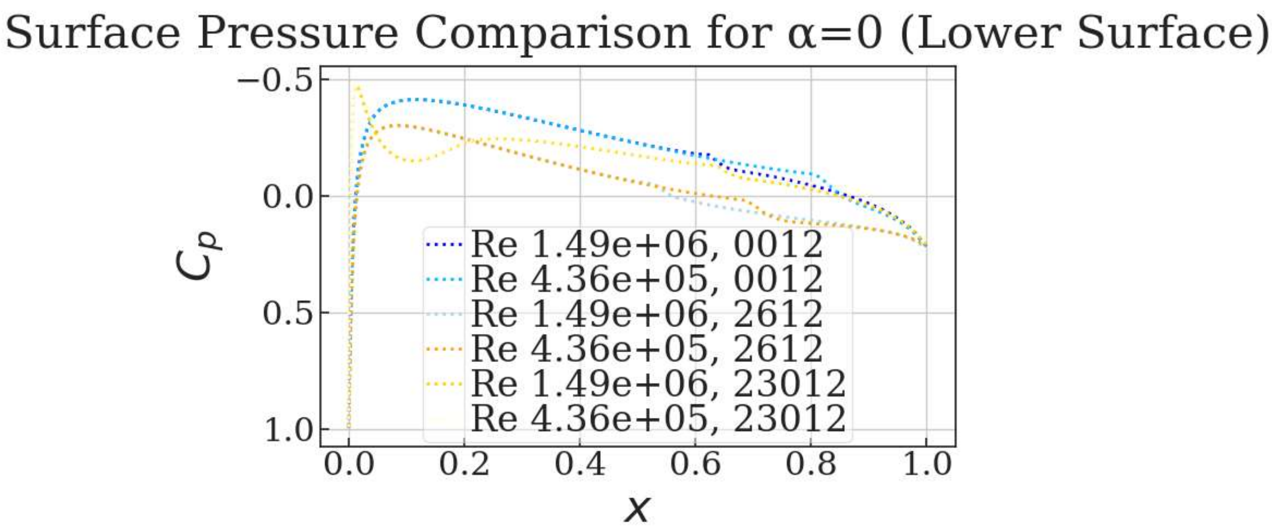
2.1 Symmetric vs Cambered Airfoil Surface Pressure Comparison

In [86]:

```
# Surface Pressure Comparison
# For Alpha = 0
# Import Data


base_path1 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca0012'
base_path2 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca2612'
base_path3 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca23012'

def load_cp_data(filename):
    return np.loadtxt(filename, skiprows=3, unpack=True, delimiter=None, usecols=(0, 2))

# Airfoil NACA 0012
```

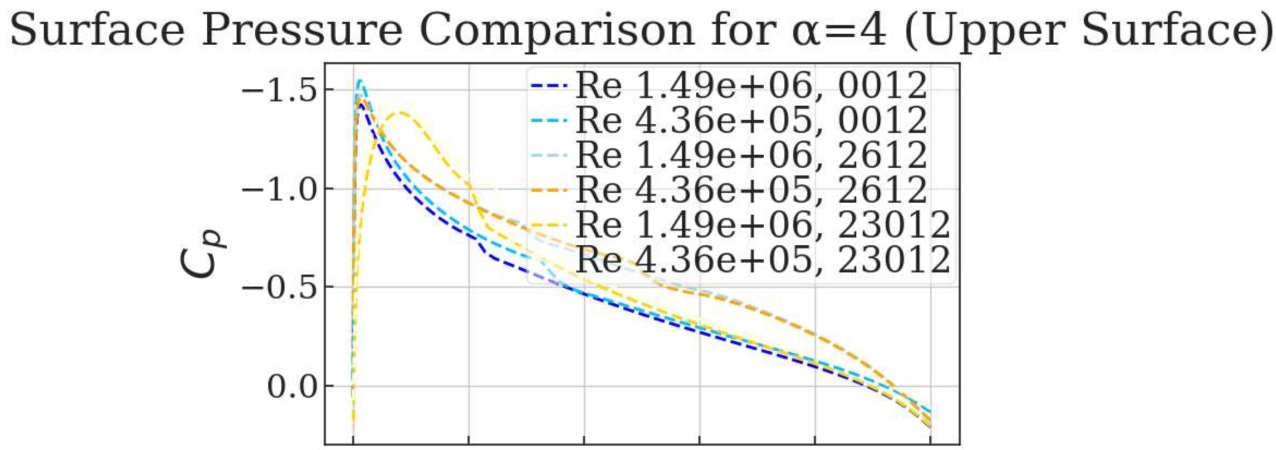
```python
x1, Cp1 = load_cp_data(os.path.join(base_path1, 'naca0012_surfCP_Re1.49e+06a0.0.dat'))
x1u, x1l = x1[:(len(x1)//2)], x1[(len(x1)//2):]
Cp1u, Cp1l = Cp1[:(len(Cp1)//2)], Cp1[(len(Cp1)//2):]

x2, Cp2 = load_cp_data(os.path.join(base_path1, 'naca0012_surfCP_Re4.36e+05a0.0.dat'))
x2u, x2l = x2[:(len(x2)//2)], x2[(len(x2)//2):]
Cp2u, Cp2l = Cp2[:(len(Cp2)//2)], Cp2[(len(Cp2)//2):]

# Airfoil NACA 2612
x3, Cp3 = load_cp_data(os.path.join(base_path2, 'naca2612_surfCP_Re1.49e+06a0.0.dat'))
x3u, x3l = x3[:(len(x3)//2)], x3[(len(x3)//2):]
Cp3u, Cp3l = Cp3[:(len(Cp3)//2)], Cp3[(len(Cp3)//2):]

x4, Cp4 = load_cp_data(os.path.join(base_path2, 'naca2612_surfCP_Re4.36e+05a0.0.dat'))
x4u, x4l = x4[:(len(x4)//2)], x4[(len(x4)//2):]
Cp4u, Cp4l = Cp4[:(len(Cp4)//2)], Cp4[(len(Cp4)//2):]

# Airfoil NACA 23012
x5, Cp5 = load_cp_data(os.path.join(base_path3, 'naca23012_surfCP_Re1.49e+06a0.0.dat'))
x5u, x5l = x5[:(len(x5)//2)], x5[(len(x5)//2):]
Cp5u, Cp5l = Cp5[:(len(Cp5)//2)], Cp5[(len(Cp5)//2):]

x6, Cp6 = load_cp_data(os.path.join(base_path3, 'naca23012_surfCP_Re4.36e+05a0.0.dat'))
x6u, x6l = x6[:(len(x6)//2)], x6[(len(x6)//2):]
Cp6u, Cp6l = Cp6[:(len(Cp6)//2)], Cp6[(len(Cp6)//2):]

# Create Subplots
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(8, 10), sharex=True)

# Upper Surface Plot
ax1.set_title('Surface Pressure Comparison for α=0 (Upper Surface)')
ax1.set_ylabel("$C_p$")

ax1.plot(x1u, Cp1u, label='Re 1.49e+06, 0012', linewidth=2, linestyle='--', color='blue')
ax1.plot(x2u, Cp2u, label='Re 4.36e+05, 0012', linewidth=2, linestyle='--', color='deepskyblue')
ax1.plot(x3u, Cp3u, label='Re 1.49e+06, 2612', linewidth=2, linestyle='--', color='lightblue')
ax1.plot(x4u, Cp4u, label='Re 4.36e+05, 2612', linewidth=2, linestyle='--', color='orange')
ax1.plot(x5u, Cp5u, label='Re 1.49e+06, 23012', linewidth=2, linestyle='--', color='gold')
ax1.plot(x6u, Cp6u, label='Re 4.36e+05, 23012', linewidth=2, linestyle='--', color='lightyellow')

ax1.invert_yaxis()  # Invert y-axis for Cp
ax1.grid(True)
ax1.legend(loc='best')

# Lower Surface Plot
ax2.set_title('Surface Pressure Comparison for α=0 (Lower Surface)')
ax2.set_xlabel("$x$")
ax2.set_ylabel("$C_p$")

ax2.plot(x1l, Cp1l, label='Re 1.49e+06, 0012', linewidth=2, linestyle='dotted', color='blue')
ax2.plot(x2l, Cp2l, label='Re 4.36e+05, 0012', linewidth=2, linestyle='dotted', color='deepskyblue')
ax2.plot(x3l, Cp3l, label='Re 1.49e+06, 2612', linewidth=2, linestyle='dotted', color='lightblue')
ax2.plot(x4l, Cp4l, label='Re 4.36e+05, 2612', linewidth=2, linestyle='dotted', color='orange')
ax2.plot(x5l, Cp5l, label='Re 1.49e+06, 23012', linewidth=2, linestyle='dotted', color='gold')
ax2.plot(x6l, Cp6l, label='Re 4.36e+05, 23012', linewidth=2, linestyle='dotted', color='lightyellow')

ax2.invert_yaxis()  # Invert y-axis for Cp
ax2.grid(True)
ax2.legend(loc='best')

# Display plots
plt.tight_layout()
plt.show()
```

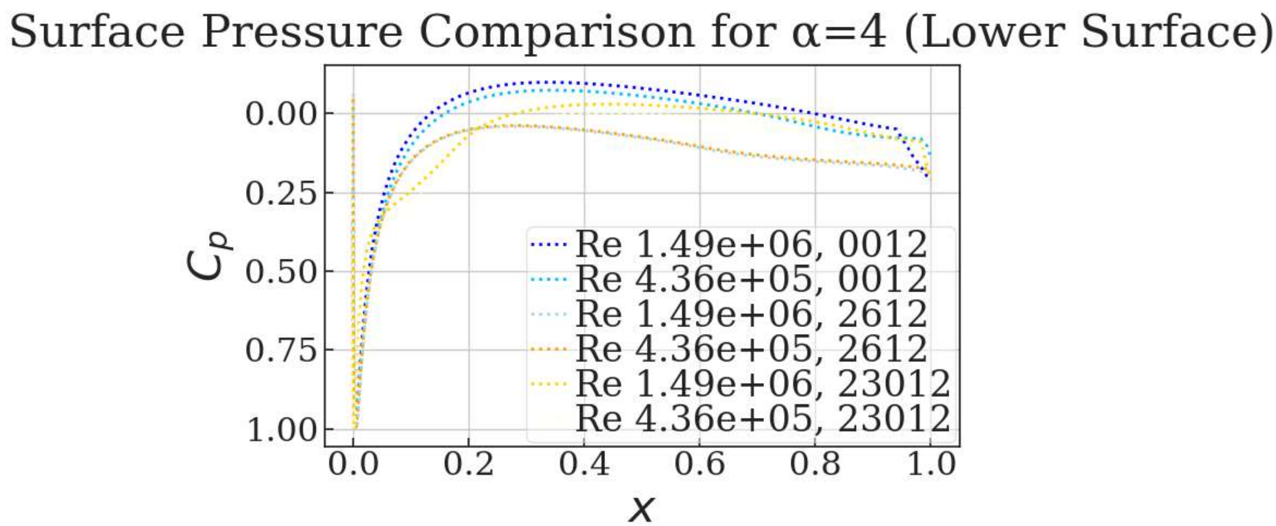




In [87]:
```
# Surface Pressure Comparison
# For Alpha = 4
# Import Data


base_path1 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca0012'
base_path2 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca2612'
base_path3 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca23012'

def load_cp_data(filename):
    return np.loadtxt(filename, skiprows=3, unpack=True, delimiter=None, usecols=(0, 2))

# Airfoil NACA 0012
x1, Cp1 = load_cp_data(os.path.join(base_path1, 'naca0012_surfCP_Re1.49e+06a4.0.dat'))
x1u, x1l = x1[:(len(x1)//2)], x1[(len(x1)//2):]
Cp1u, Cp1l = Cp1[:(len(Cp1)//2)], Cp1[(len(Cp1)//2):]

x2, Cp2 = load_cp_data(os.path.join(base_path1, 'naca0012_surfCP_Re4.36e+05a4.0.dat'))
x2u, x2l = x2[:(len(x2)//2)], x2[(len(x2)//2):]
Cp2u, Cp2l = Cp2[:(len(Cp2)//2)], Cp2[(len(Cp2)//2):]

# Airfoil NACA 2612
x3, Cp3 = load_cp_data(os.path.join(base_path2, 'naca2612_surfCP_Re1.49e+06a4.0.dat'))
x3u, x3l = x3[:(len(x3)//2)], x3[(len(x3)//2):]
Cp3u, Cp3l = Cp3[:(len(Cp3)//2)], Cp3[(len(Cp3)//2):]

x4, Cp4 = load_cp_data(os.path.join(base_path2, 'naca2612_surfCP_Re4.36e+05a4.0.dat'))
x4u, x4l = x4[:(len(x4)//2)], x4[(len(x4)//2):]
Cp4u, Cp4l = Cp4[:(len(Cp4)//2)], Cp4[(len(Cp4)//2):]

# Airfoil NACA 23012
x5, Cp5 = load_cp_data(os.path.join(base_path3, 'naca23012_surfCP_Re1.49e+06a4.0.dat'))
x5u, x5l = x5[:(len(x5)//2)], x5[(len(x5)//2):]
Cp5u, Cp5l = Cp5[:(len(Cp5)//2)], Cp5[(len(Cp5)//2):]

x6, Cp6 = load_cp_data(os.path.join(base_path3, 'naca23012_surfCP_Re4.36e+05a4.0.dat'))
```

```
x6u, x6l = x6[:(len(x6)//2)], x6[(len(x6)//2):]
Cp6u, Cp6l = Cp6[:(len(Cp6)//2)], Cp6[(len(Cp6)//2):]

# Create Subplots
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(8, 10), sharex=True)

# Upper Surface Plot
ax1.set_title('Surface Pressure Comparison for α=4 (Upper Surface)')
ax1.set_ylabel("$C_p$")

ax1.plot(x1u, Cp1u, label='Re 1.49e+06, 0012', linewidth=2, linestyle='--', color='blue')
ax1.plot(x2u, Cp2u, label='Re 4.36e+05, 0012', linewidth=2, linestyle='--', color='deepskyblue')
ax1.plot(x3u, Cp3u, label='Re 1.49e+06, 2612', linewidth=2, linestyle='--', color='lightblue')
ax1.plot(x4u, Cp4u, label='Re 4.36e+05, 2612', linewidth=2, linestyle='--', color='orange')
ax1.plot(x5u, Cp5u, label='Re 1.49e+06, 23012', linewidth=2, linestyle='--', color='gold')
ax1.plot(x6u, Cp6u, label='Re 4.36e+05, 23012', linewidth=2, linestyle='--', color='lightyellow')

ax1.invert_yaxis()  # Invert y-axis for Cp
ax1.grid(True)
ax1.legend(loc='best')

# Lower Surface Plot
ax2.set_title('Surface Pressure Comparison for α=4 (Lower Surface)')
ax2.set_xlabel("$x$")
ax2.set_ylabel("$C_p$")

ax2.plot(x1l, Cp1l, label='Re 1.49e+06, 0012', linewidth=2, linestyle='dotted', color='blue')
ax2.plot(x2l, Cp2l, label='Re 4.36e+05, 0012', linewidth=2, linestyle='dotted', color='deepskyblue')
ax2.plot(x3l, Cp3l, label='Re 1.49e+06, 2612', linewidth=2, linestyle='dotted', color='lightblue')
ax2.plot(x4l, Cp4l, label='Re 4.36e+05, 2612', linewidth=2, linestyle='dotted', color='orange')
ax2.plot(x5l, Cp5l, label='Re 1.49e+06, 23012', linewidth=2, linestyle='dotted', color='gold')
ax2.plot(x6l, Cp6l, label='Re 4.36e+05, 23012', linewidth=2, linestyle='dotted', color='lightyellow')

ax2.invert_yaxis()  # Invert y-axis for Cp
ax2.grid(True)
ax2.legend(loc='best')

# Display plots
plt.tight_layout()
plt.show()
```

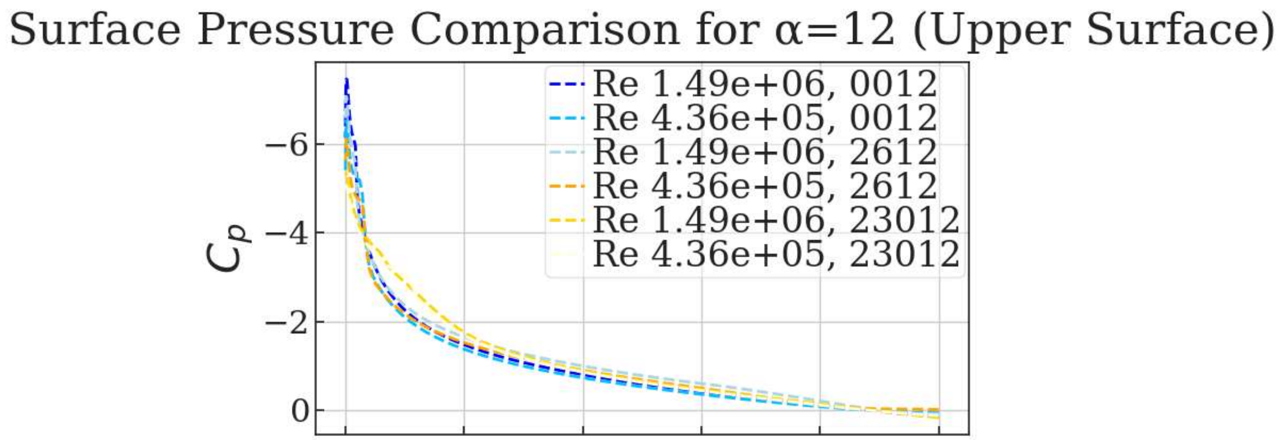


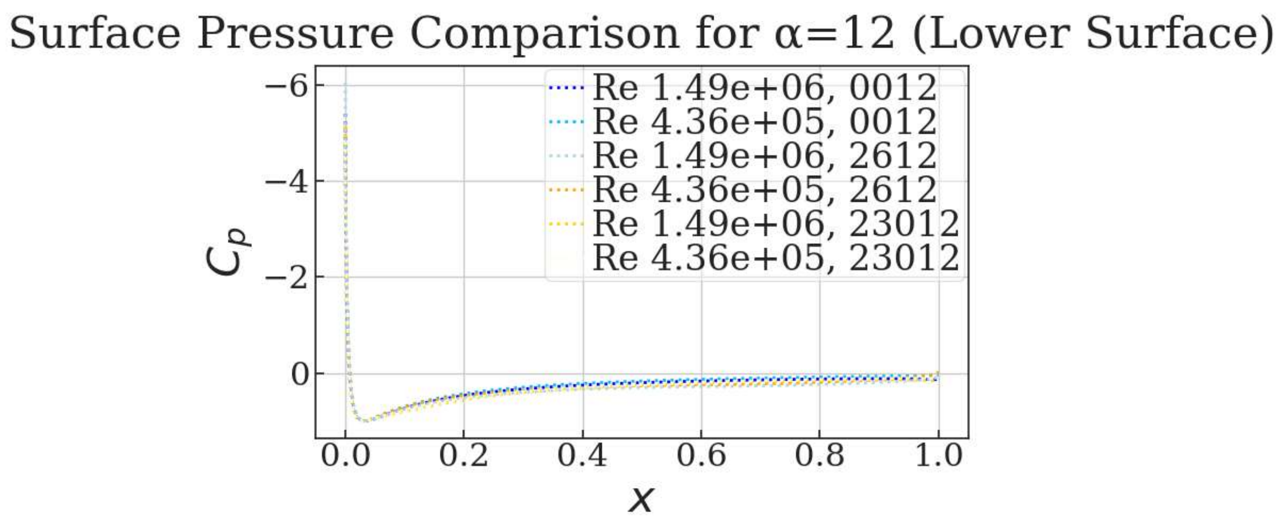


In [88]:

```python
# Surface Pressure Comparison
# For Alpha = 12
# Import Data

base_path1 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca0012'
base_path2 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca2612'
base_path3 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca23012'

def load_cp_data(filename):
    return np.loadtxt(filename, skiprows=3, unpack=True, delimiter=None, usecols=(0, 2))

# Airfoil NACA 0012
x1, Cp1 = load_cp_data(os.path.join(base_path1, 'naca0012_surfCP_Re1.49e+06a12.0.dat'))
x1u, x1l = x1[:(len(x1)//2)], x1[(len(x1)//2):]
Cp1u, Cp1l = Cp1[:(len(Cp1)//2)], Cp1[(len(Cp1)//2):]

x2, Cp2 = load_cp_data(os.path.join(base_path1, 'naca0012_surfCP_Re4.36e+05a12.0.dat'))
x2u, x2l = x2[:(len(x2)//2)], x2[(len(x2)//2):]
Cp2u, Cp2l = Cp2[:(len(Cp2)//2)], Cp2[(len(Cp2)//2):]

# Airfoil NACA 2612
x3, Cp3 = load_cp_data(os.path.join(base_path2, 'naca2612_surfCP_Re1.49e+06a12.0.dat'))
x3u, x3l = x3[:(len(x3)//2)], x3[(len(x3)//2):]
Cp3u, Cp3l = Cp3[:(len(Cp3)//2)], Cp3[(len(Cp3)//2):]

x4, Cp4 = load_cp_data(os.path.join(base_path2, 'naca2612_surfCP_Re4.36e+05a12.0.dat'))
x4u, x4l = x4[:(len(x4)//2)], x4[(len(x4)//2):]
Cp4u, Cp4l = Cp4[:(len(Cp4)//2)], Cp4[(len(Cp4)//2):]

# Airfoil NACA 23012
x5, Cp5 = load_cp_data(os.path.join(base_path3, 'naca23012_surfCP_Re1.49e+06a12.0.dat'))
x5u, x5l = x5[:(len(x5)//2)], x5[(len(x5)//2):]
Cp5u, Cp5l = Cp5[:(len(Cp5)//2)], Cp5[(len(Cp5)//2):]

x6, Cp6 = load_cp_data(os.path.join(base_path3, 'naca23012_surfCP_Re4.36e+05a12.0.dat'))
x6u, x6l = x6[:(len(x6)//2)], x6[(len(x6)//2):]
```

```
Cp6u, Cp6l = Cp6[:(len(Cp6)//2)], Cp6[(len(Cp6)//2):]

# Create Subplots
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(8, 10), sharex=True)

# Upper Surface Plot
ax1.set_title('Surface Pressure Comparison for α=12 (Upper Surface)')
ax1.set_ylabel("$C_p$")

ax1.plot(x1u, Cp1u, label='Re 1.49e+06, 0012', linewidth=2, linestyle='--', color='blue')
ax1.plot(x2u, Cp2u, label='Re 4.36e+05, 0012', linewidth=2, linestyle='--', color='deepskyblue')
ax1.plot(x3u, Cp3u, label='Re 1.49e+06, 2612', linewidth=2, linestyle='--', color='lightblue')
ax1.plot(x4u, Cp4u, label='Re 4.36e+05, 2612', linewidth=2, linestyle='--', color='orange')
ax1.plot(x5u, Cp5u, label='Re 1.49e+06, 23012', linewidth=2, linestyle='--', color='gold')
ax1.plot(x6u, Cp6u, label='Re 4.36e+05, 23012', linewidth=2, linestyle='--', color='lightyellow')

ax1.invert_yaxis()  # Invert y-axis for Cp
ax1.grid(True)
ax1.legend(loc='best')

# Lower Surface Plot
ax2.set_title('Surface Pressure Comparison for α=12 (Lower Surface)')
ax2.set_xlabel("$x$")
ax2.set_ylabel("$C_p$")

ax2.plot(x1l, Cp1l, label='Re 1.49e+06, 0012', linewidth=2, linestyle='dotted', color='blue')
ax2.plot(x2l, Cp2l, label='Re 4.36e+05, 0012', linewidth=2, linestyle='dotted', color='deepskyblue')
ax2.plot(x3l, Cp3l, label='Re 1.49e+06, 2612', linewidth=2, linestyle='dotted', color='lightblue')
ax2.plot(x4l, Cp4l, label='Re 4.36e+05, 2612', linewidth=2, linestyle='dotted', color='orange')
ax2.plot(x5l, Cp5l, label='Re 1.49e+06, 23012', linewidth=2, linestyle='dotted', color='gold')
ax2.plot(x6l, Cp6l, label='Re 4.36e+05, 23012', linewidth=2, linestyle='dotted', color='lightyellow')

ax2.invert_yaxis()  # Invert y-axis for Cp
ax2.grid(True)
ax2.legend(loc='best')

# Display plots
plt.tight_layout()
plt.show()
```



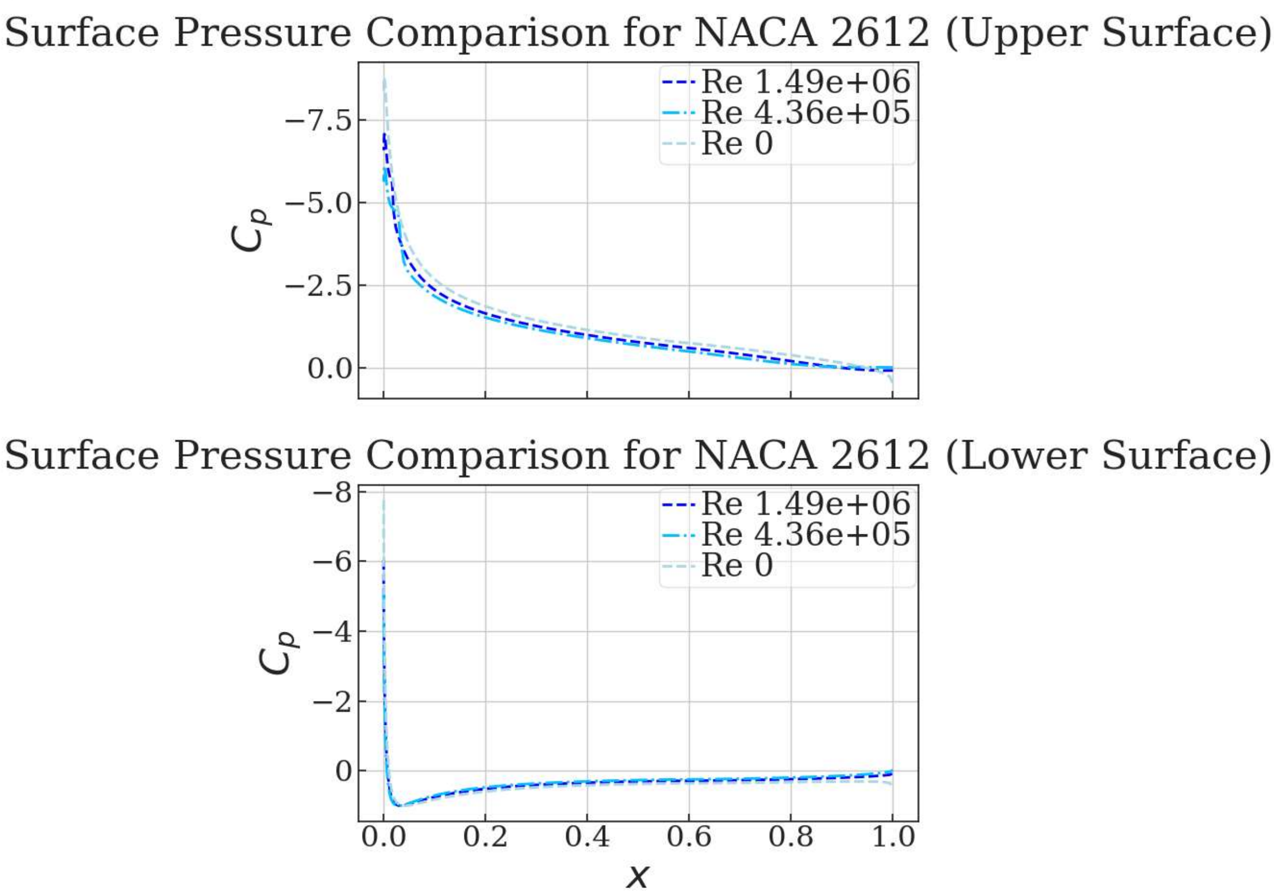


When comparing each of these airfoils at the same angle of attack, it is hard to see much of a difference between them all. Generally, the airfoils will hold almost an identical pressure profile accross the wing. However, the main difference can be seen when comparing the airfoils to themselves in their own performance at different angles of attack. It can be seen that the lower angle of attack has a lower upper and lower surface pressure than at higher angles of attack. I believe this makes sense because at lower angles of attack, less lift is generated and therefore there is a smaller pressure differential between the upper and lower surfaces of the wing. When comparing it to the higher angle of attack, the increase in pressure differential makes sense because of the increased lift.

2.2 Inviscid vs Viscous Flow Comparison

In [89]:
```
# Surface Pressure Comparison
# For NACA 2612
# Import Data


base_path = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca2612'

def load_cp_data(filename):
    return np.loadtxt(filename, skiprows=3, unpack=True, delimiter=None, usecols=(0, 2))

# Re= 1490345.149
x1, Cp1 = load_cp_data(os.path.join(base_path, 'naca2612_surfCP_Re1.49e+06a12.0.dat'))
x1u, x1l = x1[:(len(x1)//2)], x1[(len(x1)//2):]
Cp1u, Cp1l = Cp1[:(len(Cp1)//2)], Cp1[(len(Cp1)//2):]

# Re= 436034.9854
x2, Cp2 = load_cp_data(os.path.join(base_path, 'naca2612_surfCP_Re4.36e+05a12.0.dat'))
x2u, x2l = x2[:(len(x2)//2)], x2[(len(x2)//2):]
Cp2u, Cp2l = Cp2[:(len(Cp2)//2)], Cp2[(len(Cp2)//2):]

# Re= 0
x3, Cp3 = load_cp_data(os.path.join(base_path, 'naca2612_surfCP_Re0.00e+00a12.0.dat'))
x3u, x3l = x3[:(len(x3)//2)], x3[(len(x3)//2):]
Cp3u, Cp3l = Cp3[:(len(Cp3)//2)], Cp3[(len(Cp3)//2):]

# Create Subplots
```

```python
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(8, 10), sharex=True)

# Upper Surface Plot
ax1.set_title('Surface Pressure Comparison for NACA 2612 (Upper Surface)')
ax1.set_ylabel("$C_p$")

ax1.plot(x1u, Cp1u, label='Re 1.49e+06', linewidth=2, linestyle='--', color='blue')
ax1.plot(x2u, Cp2u, label='Re 4.36e+05', linewidth=2, linestyle='-.', color='deepskyblue')
ax1.plot(x3u, Cp3u, label='Re 0', linewidth=2, linestyle='--', color='lightblue')

ax1.invert_yaxis()  # Invert y-axis for Cp
ax1.grid(True)
ax1.legend(loc='best')

# Lower Surface Plot
ax2.set_title('Surface Pressure Comparison for NACA 2612 (Lower Surface)')
ax2.set_xlabel("$x$")
ax2.set_ylabel("$C_p$")

ax2.plot(x1l, Cp1l, label='Re 1.49e+06', linewidth=2, linestyle='--', color='blue')
ax2.plot(x2l, Cp2l, label='Re 4.36e+05', linewidth=2, linestyle='-.', color='deepskyblue')
ax2.plot(x3l, Cp3l, label='Re 0', linewidth=2, linestyle='--', color='lightblue')

ax2.invert_yaxis()  # Invert y-axis for Cp
ax2.grid(True)
ax2.legend(loc='best')

# Display plots
plt.tight_layout()
plt.show()
```

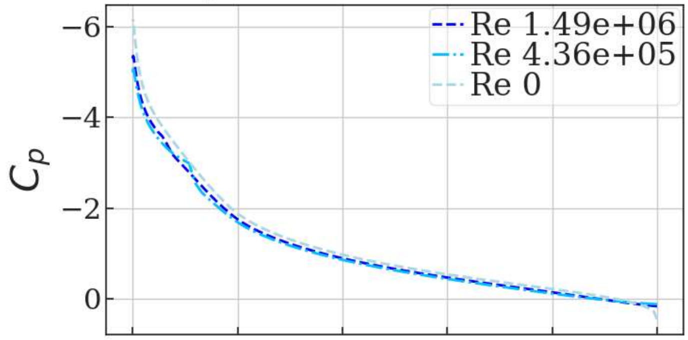


The pressure distributions for the NACA 2612 seem to be nearly identical accross the upper and lower surfaces of the airfoil. I do not see any that would be expected from a change of Reynolds number that is 6 orders of magnitude. I feel that the change is smaller than I was expecting.

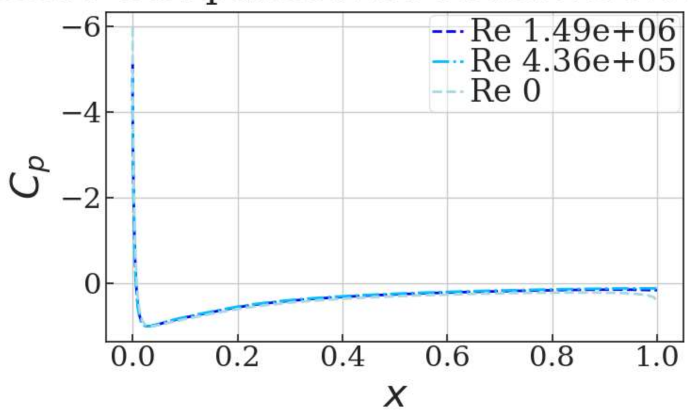
In [90]:
```python
# Surface Pressure Comparison
# For NACA 23012
# Import Data


base_path = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca23012'
```

```python
def load_cp_data(filename):
    return np.loadtxt(filename, skiprows=3, unpack=True, delimiter=None, usecols=(0, 2))

# Re= 1490345.149
x1, Cp1 = load_cp_data(os.path.join(base_path, 'naca23012_surfCP_Re1.49e+06a12.0.dat'))
x1u, x1l = x1[:(len(x1)//2)], x1[(len(x1)//2):]
Cp1u, Cp1l = Cp1[:(len(Cp1)//2)], Cp1[(len(Cp1)//2):]

# Re= 436034.9854
x2, Cp2 = load_cp_data(os.path.join(base_path, 'naca23012_surfCP_Re4.36e+05a12.0.dat'))
x2u, x2l = x2[:(len(x2)//2)], x2[(len(x2)//2):]
Cp2u, Cp2l = Cp2[:(len(Cp2)//2)], Cp2[(len(Cp2)//2):]

# Re= 0
x3, Cp3 = load_cp_data(os.path.join(base_path, 'naca23012_surfCP_Re0.00e+00a12.0.dat'))
x3u, x3l = x3[:(len(x3)//2)], x3[(len(x3)//2):]
Cp3u, Cp3l = Cp3[:(len(Cp3)//2)], Cp3[(len(Cp3)//2):]

# Create Subplots
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(8, 10), sharex=True)

# Upper Surface Plot
ax1.set_title('Surface Pressure Comparison for NACA 23012 (Upper Surface)')
ax1.set_ylabel("$C_p$")

ax1.plot(x1u, Cp1u, label='Re 1.49e+06', linewidth=2, linestyle='--', color='blue')
ax1.plot(x2u, Cp2u, label='Re 4.36e+05', linewidth=2, linestyle='-.', color='deepskyblue')
ax1.plot(x3u, Cp3u, label='Re 0', linewidth=2, linestyle='--', color='lightblue')

ax1.invert_yaxis()  # Invert y-axis for Cp
ax1.grid(True)
ax1.legend(loc='best')

# Lower Surface Plot
ax2.set_title('Surface Pressure Comparison for NACA 23012 (Lower Surface)')
ax2.set_xlabel("$x$")
ax2.set_ylabel("$C_p$")

ax2.plot(x1l, Cp1l, label='Re 1.49e+06', linewidth=2, linestyle='--', color='blue')
ax2.plot(x2l, Cp2l, label='Re 4.36e+05', linewidth=2, linestyle='-.', color='deepskyblue')
ax2.plot(x3l, Cp3l, label='Re 0', linewidth=2, linestyle='--', color='lightblue')

ax2.invert_yaxis()  # Invert y-axis for Cp
ax2.grid(True)
ax2.legend(loc='best')

# Display plots
plt.tight_layout()
plt.show()
```

Surface Pressure Comparison for NACA 23012 (Upper Surface)



Surface Pressure Comparison for NACA 23012 (Lower Surface)



The pressure distributions for the NACA 23012 seem to be nearly identical accross the upper and lower surfaces of the airfoil. I do not see any that would be expected from a change of Reynolds number that is 6 orders of magnitude. I feel that the change is smaller than I was expecting.

2.3 Airfoil and Drag Polars

In [91]:
```
import pandas as pd
import sys
import pyxfoil

foil = '23012'
naca = True
Re   = 436034.9854
alfs = np.linspace(-12, 12, 50) #Angles of Attack
print("Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...\n")
pyxfoil.GetPolar(foil, naca, alfs, Re, SaveCP=True, quiet=True)


foil = '2612'
naca = True
Re   = 436034.9854
alfs = np.linspace(-12, 12, 50) #Angles of Attack
print("Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...\n")
pyxfoil.GetPolar(foil, naca, alfs, Re, SaveCP=True, quiet=True)
```

```
Running XFOIL/pyxfoil and saving data in 'Data/naca0012' folder...

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```
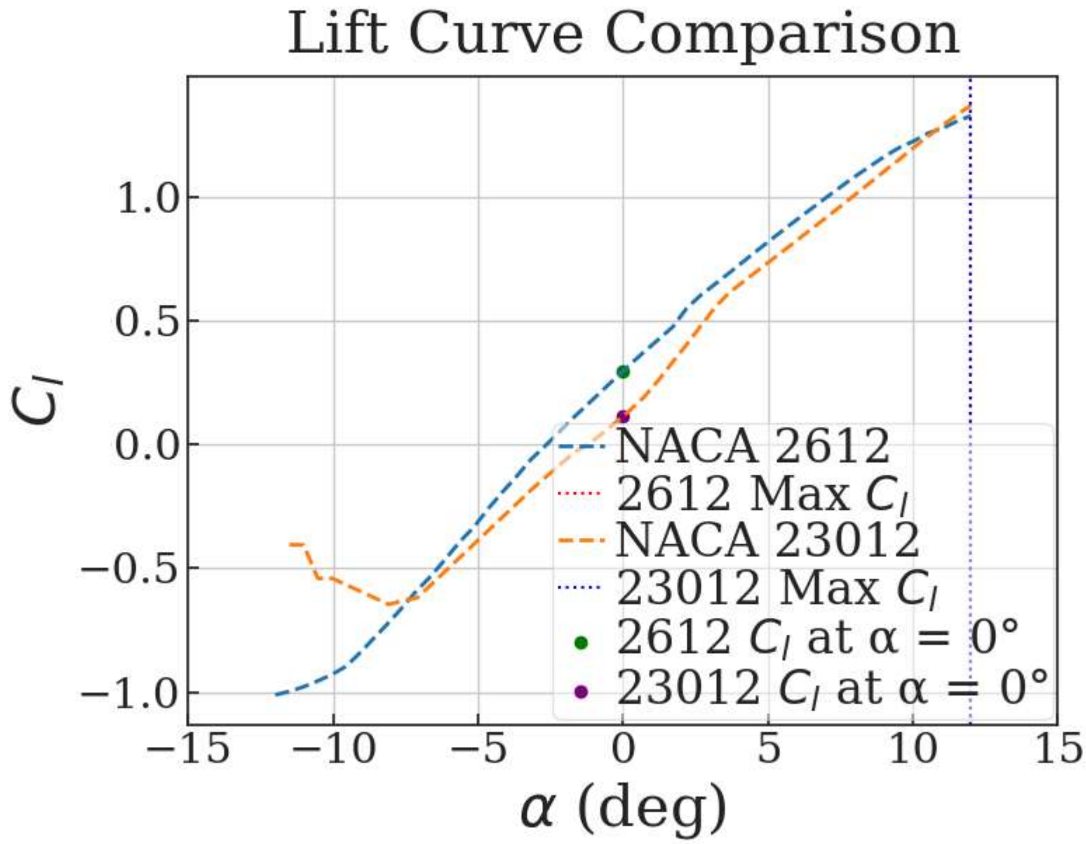
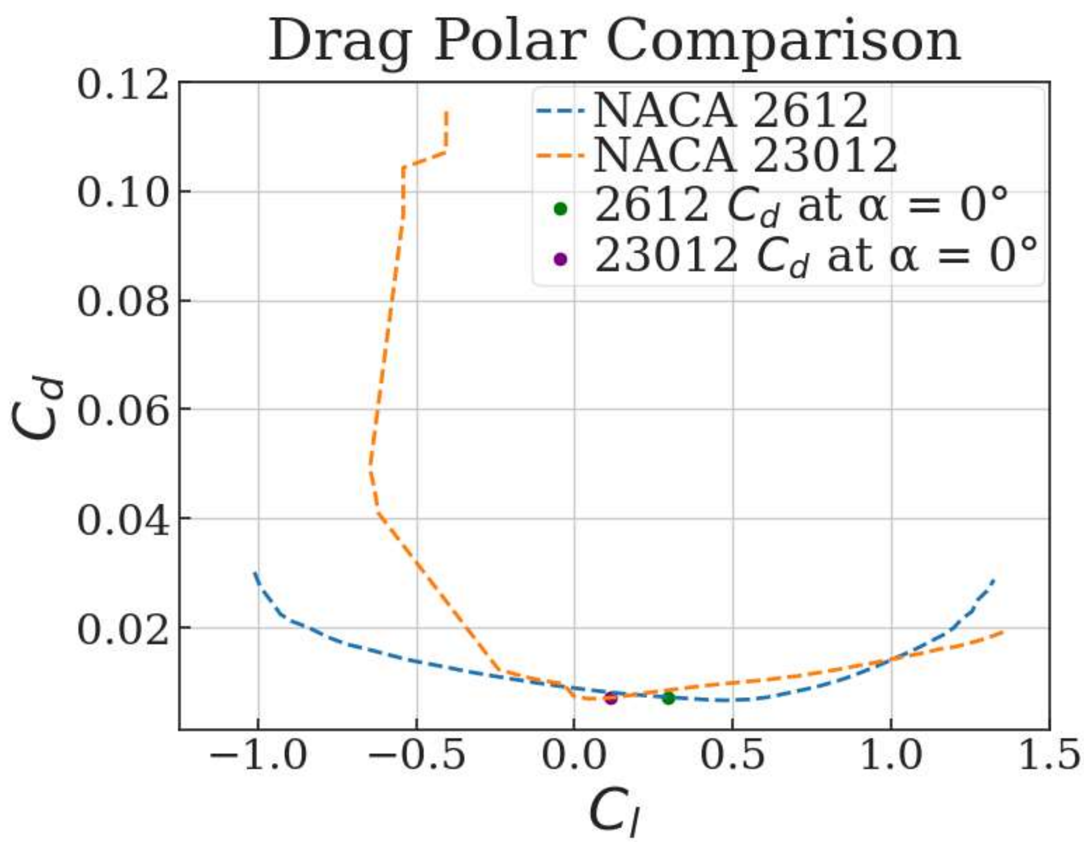

Out[91]: `<pyxfoil.Xfoil at 0x23e19a494f0>`

In [92]:
```
#Re = Case 1


filename_2612 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca2612\\naca2612_polar_Re4.36e+05a-
Alpha4, Cl4, Cd4 = np.loadtxt(filename_2612, skiprows=12, unpack=True, usecols=(0, 1, 2))
filename_23012 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca23012\\naca23012_polar_Re4.36e+0
Alpha5, Cl5, Cd5 = np.loadtxt(filename_23012, skiprows=12, unpack=True, usecols=(0, 1, 2))

# Find maximum and minimum Cl values for NACA 2612
max_Cl4_index = np.argmax(Cl4)
```

```python
max_Cl4 = Cl4[max_Cl4_index]
alphamaxCl4 = Alpha4[max_Cl4_index]

min_Cl4_index = np.argmin(Cl4)
min_Cl4 = Cl4[min_Cl4_index]
alphaminCl4 = Alpha4[min_Cl4_index]

# Find maximum and minimum Cl values for NACA 23012
max_Cl5_index = np.argmax(Cl5)
max_Cl5 = Cl5[max_Cl5_index]
alphamaxCl5 = Alpha5[max_Cl5_index]

min_Cl5_index = np.argmin(Cl5)
min_Cl5 = Cl5[min_Cl5_index]
alphaminCl5 = Alpha5[min_Cl5_index]

# Stall angle of attack = maximum lift coefficient
stall_angle_Cl4 = alphamaxCl4
stall_angle_Cl5 = alphamaxCl5

# Linear interpolation to find the y-intercept (Cl and Cd at alpha = 0)
alphazeroCl4 = np.interp(0, Alpha4, Cl4)
alphazeroCd4 = np.interp(0, Alpha4, Cd4)

alphazeroCl5 = np.interp(0, Alpha5, Cl5)
alphazeroCd5 = np.interp(0, Alpha5, Cd5)

# Plot Lift Curves
plt.figure(figsize=(8, 6))
plt.title('Lift Curve Comparison')
plt.xlabel("$\\alpha$ (deg)")
plt.ylabel("$C_l$")

plt.plot(Alpha4, Cl4, label='NACA 2612', linewidth=2, linestyle='--')
plt.axvline(alphamaxCl4, color='red', linestyle=':', label=f'2612 Max $C_l$')

plt.plot(Alpha5, Cl5, label='NACA 23012', linewidth=2, linestyle='--')
plt.axvline(alphamaxCl5, color='blue', linestyle=':', label=f'23012 Max $C_l$')

# Mark interpolated zero angle of attack for both airfoils
plt.scatter(0, alphazeroCl4, color='green', label=f'2612 $C_l$ at α = 0°')
plt.scatter(0, alphazeroCl5, color='purple', label=f'23012 $C_l$ at α = 0°')

plt.grid(True)
plt.xlim([-15, 15])
plt.legend(loc='best')
plt.show()

# Plot Drag Polars
plt.figure(figsize=(8, 6))
plt.title('Drag Polar Comparison')
plt.xlabel("$C_l$")
plt.ylabel("$C_d$")

plt.plot(Cl4, Cd4, label='NACA 2612', linewidth=2, linestyle='--')
plt.plot(Cl5, Cd5, label='NACA 23012', linewidth=2, linestyle='--')

# Mark interpolated zero angle of attack drag for both airfoils
plt.scatter(alphazeroCl4, alphazeroCd4, color='green', label=f'2612 $C_d$ at α = 0°')
plt.scatter(alphazeroCl5, alphazeroCd5, color='purple', label=f'23012 $C_d$ at α = 0°')

plt.grid(True)
plt.xlim([-1.25, 1.5])
plt.legend(loc='best')
plt.show()

# Print key information
print(f"NACA 2612 - Maximum C_l: {max_Cl4:.4f} at α = {alphamaxCl4:.2f}°")
print(f"NACA 23012 - Maximum C_l: {max_Cl5:.4f} at α = {alphamaxCl5:.2f}°")
print(f"NACA 2612 - Stall angle: {stall_angle_Cl4:.2f}°")
print(f"NACA 23012 - Stall angle: {stall_angle_Cl5:.2f}°")
print(f"NACA 2612 - Zero α : C_l = {alphazeroCl4:.4f}, C_d = {alphazeroCd4:.4f}")
print(f"NACA 23012 - Zero α : C_l = {alphazeroCl5:.4f}, C_d = {alphazeroCd5:.4f}")
```






```
NACA 2612 - Maximum C_l: 1.3260 at α = 12.00°
NACA 23012 - Maximum C_l: 1.3686 at α = 12.00°
NACA 2612 - Stall angle: 12.00°
NACA 23012 - Stall angle: 12.00°
NACA 2612 - Zero α : C_l = 0.2960, C_d = 0.0072
NACA 23012 - Zero α : C_l = 0.1142, C_d = 0.0071
```

2.4 Force and Moment Integration

In [109…

```
import numpy as np
import matplotlib.pyplot as plt
import os

# Load data from file
base_path3 = 'C:\\Users\\ephra\\OneDrive\\Desktop\\PythonScripts\\EAE127\\.git\\Data\\naca23012'

def load_cp_data(filename):
    return np.loadtxt(filename, skiprows=12, unpack=True, delimiter=None, usecols=(0, 1, 2))

def calculate_aerodynamic_properties(filename):
    # Load data
    x, y, Cp = load_cp_data(filename)

    # Chord length
    c = 1.0

    # Calculate lower and upper surface pressure coefficients
    Cp_l = Cp[:len(Cp)//2]   # Lower surface (first half)
    Cp_u = Cp[len(Cp)//2:]   # Upper surface (second half)

    # Ensure both Cp_l and Cp_u are the same length
    min_length = min(len(Cp_l), len(Cp_u))
    Cp_l = Cp_l[:min_length]
    Cp_u = Cp_u[:min_length]
    x = x[:min_length]  # Adjust x to match the new length
    y_l = y[:min_length]  # Lower surface y values
    y_u = y[len(y)//2:len(y)//2 + min_length]  # Upper surface y values

    # Integrate for Cn
    Cn = (1 / c) * np.trapz(Cp_l - Cp_u, x)

    # Calculate surface slopes using finite differences
    dz_dx_u = np.gradient(y_u, x)  # Upper surface
    dz_dx_l = np.gradient(y_l, x)  # Lower surface

    # Calculate Ca
    Ca = (1 / c) * np.trapz(dz_dx_u * Cp_u - dz_dx_l * Cp_l, x)

    # Calculate Cm,LE
    Cm_LE = (1 / c) * 0.5 * np.trapz(Cp_u - Cp_l, x)

    # Calculate lift coefficient Cl and drag coefficient Cd
    alpha = np.radians(12.0)  # Convert angle to radians
    Cl = Cn * np.cos(alpha) - Ca * np.sin(alpha)
    Cd = Cn * np.sin(alpha) + Ca * np.cos(alpha)

    # Moment about the quarter chord
    Cm_c = Cm_LE + Cl * (c / 4)

    return Cl, Cd, Cm_c, x, Cp

# File paths for three Reynolds numbers
filenames = [
    os.path.join(base_path3, 'naca23012_surfCP_Re1.49e+06a12.0.dat'),
    os.path.join(base_path3, 'naca23012_surfCP_Re4.36e+05a12.0.dat'),
    os.path.join(base_path3, 'naca23012_surfCP_Re0.00e+00a12.0.dat'),
]

# XFOIL results manual input
xf_results = [
    (1.4203, 0.01436, -0.0059),  # Re1.49e+06
    (1.3686, 0.01945, 0.0022),   # Re4.36e+05
    (1.5747, 0.0, -0.0311),      # Re0.00e+00
]

# Process each dataset and store results
results = []
for i, filename in enumerate(filenames):
    Cl, Cd, Cm_c, x, Cp = calculate_aerodynamic_properties(filename)
    results.append((Cl, Cd, Cm_c, x, Cp, xf_results[i]))

# Print results
print(f"NACA 23012 at α = 12°")
print("Re\t\tCl\t\tCd\t\tCm")
for i, (Cl, Cd, Cm_c, x, Cp, (Cl_xf, Cd_xf, Cm_xf)) in enumerate(results):
```

```
    print(f"Re{i+1}\t{Cl_xf:.5f}\t{Cd_xf:.5f}\t{Cm_xf:.5f}")
    print(f"Integration\t{Cl:.5f}\t{Cd:.5f}\t{Cm_c:.5f}")
```

```
NACA 23012 at α = 12°
Re              Cl              Cd              Cm
Re1     1.42030 0.01436 -0.00590
Integration     1.40337 -0.00253        -0.33525
Re2     1.36860 0.01945 0.00220
Integration     1.37307 -0.00168        -0.32809
Re3     1.57470 0.00000 -0.03110
Integration     1.48796 -0.00765        -0.35494
```

My integrated values seem pretty close to the actual. This took a while to figure out and a lot of my previous attempts did not even come close. I found that my biggest error was finding that numpy takes it's angle values in radians and not degrees. Once I fixed that and I equalized the lengths of the upper and lower portion, I finally got something close to the Xfoil data. I think these differences now may be from trunkation errors that come from the numpy trapz numerical integration method compared to that of xfoil. Additionally, the limited number of data points in the xfoil data may have decreased the accuracy of the output.